HTTP/1 vs HTTP/2 vs HTTP/3

HTTP/1

- Text-based messages
- No protocol-level multiplexing
- Multiple connections often used

HTTP/2

- Binary framing over
- HPACK header compression
- TCP Multiplexed request streams

HTTP/3

- HTTP semantics over QUIC (use of UDP and TLS 1.3)
- Independent QUIC streams
- QPACK header compression

✓ HTTP/2 and HTTP/3 are compatible with the HTTP/1 semantics, **only the delivery changes**

HTTP 1

Every HTTP request/response is sent over its own TCP request

But with **`keep-alive`** header, several request/response can be sent sequentially through the same TCP stream

Problem: no parallelism unless opening several TCP streams